An LLM-Based Approach to Improving Throughput and Reducing Collisions in IEEE 802.11 Wireless Networks

Guan-Sheng Huang[1], Chih-Heng Ke[1]*, Chong-Yi Yang[2], and Yeong-Sheng Chen[3]

[1] Department of Computer Science and Information Engineering, National Quemoy University, Kinmen, Taiwan, ROC
[2] CHT Security Co., Ltd., Taipei, Taiwan, ROC
[3] Department of Computer Science, National Taipei University of Education, Taipei, Taiwan, ROC

Abstract. We propose DCFLLM (Distributed Coordination Function Large Language Model), a novel large language model-based framework designed for intelligent contention window (CW) optimization in IEEE 802.11 wireless LANs. Unlike reinforcement learning (RL) methods that require retraining across different environments, DCFLLM leverages pre-trained language models to infer network dynamics and recommend CW adjustments directly. By integrating protocol knowledge from technical documents and empirical data, the model achieves adaptive, zero-shot decision-making without relying on trial-and-error learning. DCFLLM supports multi-objective optimization including collision rate minimization and throughput enhancement and adapts to varying node densities and traffic loads. It can also interface with federated or hierarchical control systems for scalable deployment. Experimental simulations confirm that DCFLLM significantly improves quality of service (QoS), reduces collision rates, and eliminates the overhead of RL training. This work pioneers a new class of intelligent, training-free and interpretable solutions for proactive wireless network control.

Keywords: IEEE 802.11 · Wireless network optimization · LLM (Large Language Model) · RL (Reinforcement Learning) · QoS · Intelligent network management.

1 Introduction

In modern wireless communication systems, optimizing bandwidth utilization and ensuring efficient network operation have become critical challenges. IEEE 802.11, as the foundation of wireless LANs, is essential for maintaining data transmission reliability and quality of service (QoS). However, the rapid growth of devices and applications such as IoT and extended reality (XR) has intensified network pressure, leading to increased collision rates and reduced throughput

* Corresponding author: Chih-Heng Ke (smallko@gmail.com)

in high-density scenarios. Existing mechanisms like binary exponential backoff (BEB) offer basic contention control but struggle under dynamic or congested environments, resulting in suboptimal channel usage and fairness issues. Reinforcement learning (RL) [9] has been proposed to address these problems through adaptive parameter tuning, but its reliance on long training times and sensitivity to environmental changes limits its practicality.

Moreover, the contention window (CW) size directly influences collision probability. The probability that two stations select the same backoff time is inversely related to the size of the CW. Increasing CW reduces the chance of collision; however, excessively large CW values introduce unnecessary delay. Therefore, identifying a balanced CW setting is essential for achieving both low latency and minimal contention loss.

To overcome these constraints, we explore a novel approach that integrates large language models (LLMs) with IEEE 802.11 resource management. Unlike RL-based methods, LLMs can leverage prior knowledge to make real-time, context-aware decisions without retraining. Trained on wireless protocol documents and empirical datasets, an LLM can analyze network states, diagnose contention problems, and recommend CW [3] and backoff configurations.

The central research question of this study is whether a general-purpose Large Language Model (LLM) can effectively substitute for a specialized reinforcement learning agent, such as a Deep Q-Network (DQN), in the complex task of dynamic wireless network management. Focusing on multi-node environments, this work experimentally evaluates the LLM's ability to enhance key performance metrics like throughput by minimizing collision rates. By directly comparing the LLM's prompt-based decision-making against a trained DQN agent, we seek to demonstrate that intelligent and interpretable network optimization can be achieved without the significant training overhead, data collection, and environmental sensitivity that characterize conventional RL-based techniques. Ultimately, this study investigates if LLMs offer a more practical and flexible path toward intelligent resource management in IEEE 802.11 systems.

To address the aforementioned issues in a systematic manner, this paper is organized as follows. Section 2 lays the theoretical foundation by reviewing the relevant background and discussing key concepts and related work essential to understanding the problem domain. Section 3 introduces the proposed approach in detail, including the design rationale, system architecture, and implementation methodology. Section 4 presents a comprehensive experimental evaluation to assess the performance and effectiveness of the proposed method, supported by both quantitative results and qualitative analysis. Section 5 concludes the paper by summarizing the key findings, highlighting the contributions, and outlining potential directions for future research.

1.1 DCFLLM

The proposed framework, named DCFLLM, describes a novel direction in wireless network optimization by utilizing pretrained large language models (LLMs)

as reinforcement learning agents. Rather than relying on traditional policy training, DCFLLM is prompted to act as a decision-maker based on real-time network conditions. For example, a representative prompt might state: "You are playing the role of a reinforcement learning agent. I will provide you with the observed throughput and collision rate. You need to define the observations, the reward function, and determine the suitable contention window value as an action to maximize system throughput. When the LLM provides the suitable contention window value, it will output the value in JSON format." The model can directly select optimal contention window (CW) values without additional training or reliance on external modules. Through local or cloud-based deployment at access points or central controllers, DCFLLM enables intelligent and adaptive control of wireless parameters in IEEE 802.11 networks.

Recent literature supports the feasibility of applying LLMs to wireless network control. For example, Wireless LLM demonstrates that LLMs can infer resource management decisions, such as power control and frequency selection, through prompt-based interaction with real-time network statistics [11]. Similarly, LLM4Wireless explores the zero-shot capability of LLMs to perform wireless control tasks without gradient-based training or policy iteration, showcasing rapid adaptability across novel scenarios [6]. LLMs can interpret network state and generate suitable MAC-layer parameters like CWmin or AIFS using simple prompt designs [12]. Finally, AutoMAC introduces a framework where LLMs automate MAC protocol design, offering flexible contention window tuning and slot duration control based entirely on prompt reasoning [5]. These works suggest that LLMs, when appropriately prompted, can reason over wireless context and serve as an alternative to traditional RL-based control systems. DCFLLM inherits this prompt-driven architecture to optimize IEEE 802.11 contention behavior in a low-latency, deployable fashion.

2 Related works

Recent advancements in AI-driven techniques, particularly Reinforcement Learning (RL) [9], have emerged as promising solutions for optimizing the Medium Access Control (MAC) layer by dynamically adjusting control parameters in diverse WLAN conditions. Traditional backoff algorithms, such as Binary Exponential Backoff (BEB), often suffer from poor fairness and high collision rates under heavy load. To address these limitations, various improved mechanisms have been proposed, including Exponential Increase Exponential Decrease (EIED), Linear Increase Linear Decrease (LILD), and Self-Tuning Exponential and Linear (SETL) algorithms, which adapt contention windows (CW) based on traffic conditions [4, 1].

Building on these, Deep Reinforcement Learning (DRL) techniques have been applied to enable intelligent CW optimization. For instance, Deep Q-Networks (DQN) and Deep Deterministic Policy Gradient (DDPG) have been employed to adapt CW values according to observed collision rates and throughput performance [9, 7]. These models, trained over time in simulated wireless environments,

can dynamically adjust parameters to reduce collisions and enhance quality of service (QoS). However, DRL-based approaches still face challenges such as unstable learning, overestimation bias, and high training overhead, which limits real-time deployment in fast-changing networks [2, 10].

In contrast, recent research has explored Large Language Models (LLMs) as an emerging alternative. The Wireless LLM framework demonstrates that LLMs can be guided through prompt engineering and retrieval-augmented generation (RAG) to perform adaptive network configuration [10, 11], fault diagnosis, and parameter reasoning. By leveraging external knowledge sources such as standards documents or simulation code, LLMs offer zero-shot adaptability and interpretability without traditional learning cycles. These developments highlight the potential of LLMs as flexible, inference-driven agents in wireless network control, complementing or replacing training-intensive RL approaches.

Tong et al. [8] proposed Wireless Agent, a Large Language Model (LLM)-based framework designed for intelligent wireless network control. Inspired by human cognitive architectures, Wireless Agent integrates a perception-memory-planning-action cycle to manage network resource slicing. Their results demonstrate that the system achieves a 44.4% improvement in bandwidth utilization compared to prompt-only methods, while approaching the performance of rule-based optimal configurations in terms of energy efficiency and spectrum allocation.

3 System model

To achieve adaptive and intelligent medium access control in IEEE 802.11 WLANs, we propose the integration of a Large Language Model (LLM) into the system architecture to act as a centralized decision-making agent. The overall framework is illustrated in Fig. 1, DCFLLM-assisted WLAN architecture. The AP observes collision probabilities from STAs and queries the LLM to determine an optimal CW. This CW is chosen from a predefined set of discrete values: {32, 64, 128, 256, 512, 1024}, based on the current traffic conditions in 802.11b. The updated CW is broadcast to all STAs via beacon frames for coordinated access control.

In this architecture (Fig. 1), the Access Point (AP) continuously gathers real-time network statistics, including throughput and collision rate, from all connected stations (STAs). These statistics are periodically sent to an LLM, which may reside at the edge or in the cloud. The LLM, prompted with a structured instruction, interprets the network condition and selects an optimized value for the CW which based on the current traffic conditions. The selected CW is then encapsulated in JSON format and broadcast to all STAs via standard IEEE 802.11 beacon frames. Upon receiving the update, each STA aligns its initial backoff behavior accordingly. Once applied, the CW is used as the starting point for the standard binary exponential backoff procedure. Specifically, after a transmission failure (collision), the contention window doubles until reaching a fixed maximum ($CWmax = 1024$), consistent with the original DCF behavior. Con-

versely, upon a successful transmission, the contention window is reset to the most recent CW value determined by the LLM.

This centralized control framework eliminates the need for distributed learning or coordination among STAs and avoids the prolonged training cycles typically required in reinforcement learning (RL)-based solutions. By offloading policy computation to the LLM, this approach enables dynamic adaptability, reduces protocol complexity, and requires only minimal changes to existing devices.

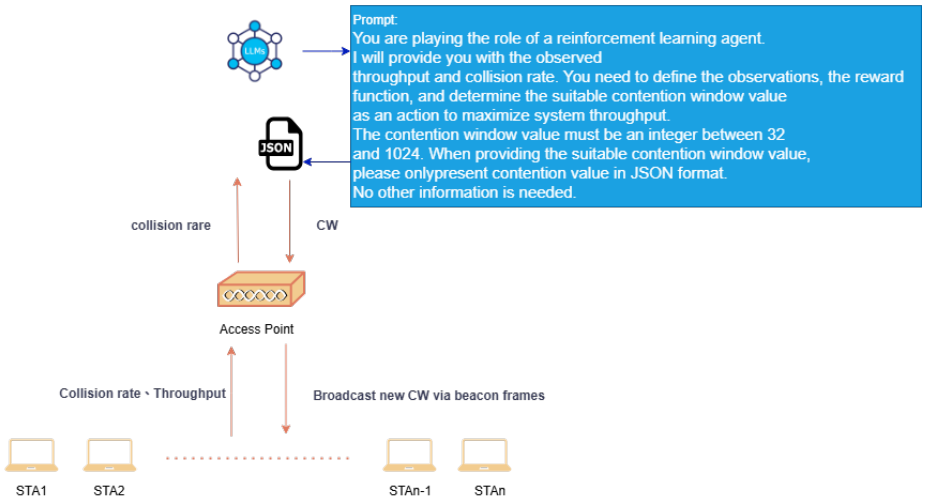

Fig. 1. DCFLLM-assisted wireless LAN architecture

3.1 Large Language Model

In this work, we utilize advanced large language models (LLMs) with strong reasoning and decision-making capabilities as the core of the DCFLLM framework. Specifically, we experiment with models such as Gemini 2.5 Pro and DeepSeek R1 30B, both of which are known for their ability to perform complex reasoning tasks, structured output generation, and few-shot generalization. These capabilities are essential for interpreting dynamic wireless network states and producing optimal configuration decisions through natural language prompts. Unlike traditional neural networks or policy-based agents that require environment-specific training, these state-of-the-art LLMs can perform zero-shot or prompt-based decision-making, making them well-suited for wireless network control scenarios where adaptability and real-time response are critical. Their inherent understanding of language and structured instructions enables them to simulate reinforcement learning behavior without explicit training.

In the context of DCFLLM, these models are deployed either locally at edge servers or remotely via cloud APIs. Upon receiving input metrics such as throughput and collision rate, the LLM is prompted to determine the optimal contention window value for IEEE 802.11 WLAN operation. Thanks to their strong internal representation and context awareness, models like Gemini and DeepSeek can infer patterns, estimate network congestion, and generate decisions that closely match or exceed the performance of traditional RL agents. By leveraging such powerful LLMs, DCFLLM achieves a balance between intelligence, flexibility, and deployment efficiency—eliminating the need for custom training pipelines while still enabling fine-grained control over dynamic wireless environments.

3.2 Prompt Template

In this study, we design a prompt template to simulate the behavior of a reinforcement learning agent for network optimization. The agent receives observations of system performance, specifically throughput and collision rate, and is required to define these observations, formulate an appropriate reward function, and determine an action.

The prompt used in our experiments is: "You are playing the role of a reinforcement learning agent. I will provide you with the observed throughput and collision rate. You need to define the observations, the reward function, and determine the suitable contention window value as an action to maximize system throughput. The contention window value must be an integer between 32 and 1024. When providing the suitable contention window value, please only present contention value in JSON format. No other information is needed."

3.3 Design Challenges and Implementation Considerations

Incorporating large language models (LLMs) into real-time decision-making for wireless networks presents several practical challenges, especially when considering various deployment methods and model sizes. For example, cloud-based models like Gemini 2.5 Pro provide advanced reasoning abilities but often experience notable response delays due to internal processing and network communication. Instead of discarding the Gemini model, our system let its CW adjustment take effect in the next control cycle instead, ensuring that high-quality decisions are integrated without compromising the system's timing stability. This approach works well because wireless traffic patterns tend to be temporally correlated, so slightly delayed decisions remain relevant and useful.

Conversely, locally hosted models such as DeepSeek R1 30B offer quicker inference times and remove reliance on external APIs, but they are more prone to reasoning errors like repetitive responses or failure to converge especially when prompts lack clear instructions or defined roles. To mitigate this, we design subjective, role-specific prompts that clearly specify the LLM's task, decision

context, and action limitations. By framing the prompt to mimic a reinforcement learning agent's goal—such as maximizing throughput within a set CW range—we greatly enhance the model's reliability and effectiveness.

Overall, the mechanism in the proposed DCFLLM framework can balance the model performance, inference speed, and system stability. The framework shows that with appropriate architectural and prompt engineering adjustments, LLMs can function as effective controllers in wireless networks without the need for traditional training processes.

4 Performance Evaluation

In this section, we evaluate the performance of our proposed DCFLLM framework. We compare its effectiveness against both the standard IEEE 802.11 Distributed Coordination Function (DCF) and a conventional Deep Q-Network (DQN) based reinforcement learning agent. The evaluation focuses on two key metrics: aggregate throughput and collision rate, under varying network loads (i.e., number of nodes).

4.1 Experimental environment

The performance evaluation was conducted using a custom-developed simulator that models an IEEE 802.11b WLAN environment. The number of stations (STAs) varies from 10 to 50 to simulate different levels of network congestion. We compare the performance of the following four schemes:

- **Gemini**: Our proposed architecture using the cloud-based Gemini 2.5 Pro model to determine the CW.
- **DeepSeek R1**: Our proposed architecture using the locally-deployed DeepSeek R1 model, executed via the Ollama runtime environment for efficient on-device inference.
- **DQN**: A state-of-the-art reinforcement learning agent that is pre-trained to dynamically select the optimal CW value. This represents a conventional, highly specialized AI-based approach.
- **DCF**: The standard IEEE 802.11 MAC protocol with a fixed CW value. This serves as our baseline to demonstrate the benefits of adaptive contention control.

4.2 Performance Metrics

To comprehensively assess the performance, we use the two metrics including aggregate throughput and collision rate.

Aggregate Throughput (bps): This measures the total number of successfully transmitted data bits per second for all STAs in the network. It is the primary indicator of network efficiency and user-perceived performance. The throughput is calculated by Equation (1), In this formula, T_{suc} represents the

cumulative time of all successful data transmissions. This value is multiplied by the data rate (the physical transmission rate, e.g., in Mbps) to find the total number of bits successfully delivered. This total bit count is then divided by the total observation time, t, to normalize the result into a rate of bits per second (bps).

$$Aggregate\ Throughput\ (bps) = \frac{T_{suc} \times datarate}{t} \tag{1}$$

Collision Rate (%): This metric reflects the proportion of channel time wasted due to packet collisions. A high collision rate indicates severe contention among STAs and directly reflects the effectiveness of the contention resolution mechanism. It is calculated using Equation (2), $N_{col,t}$ is the total count of the channel was occupied by transmissions that resulted in a collision. The denominator, $N_{suc,t} + N_{col,t}$, represents the total count of the packet which channel was busy (either with successful transmissions or collisions). Therefore, the formula calculates the fraction of busy channel time that was unproductive due to collisions, and this fraction is expressed as a percentage.

$$Collision\ Rate(\%) = \frac{N_{col,t}}{N_{suc,t} + N_{col,t}} \cdot 100\% \tag{2}$$

Table 1. Simulation Parameters

Parameter	Value
Slot time	20μs
DIFS	50μs
SIFS	10μs
CWmin / CWmax	32 / 1024

4.3 Evaluation

The experimental results, visualized in Fig. 2, present a comprehensive comparative performance analysis of four distinct methods: gemini2.5pro, deepseek_rl, DCF, and DQN, across varying numbers of network nodes. This single figure illustrates two key performance indicators: Collision Rate and Aggregate Throughput.

In terms of collision rate, as depicted in Fig. 2 (left plot), DQN consistently exhibits the lowest values, highlighting its superior ability to manage network access and minimize data collisions. Deepseek_rl, on the other hand, shows an increasing collision rate as the number of nodes grows, while gemini2.5pro experiences a significant rise in its collision rate with network expansion. DCF consistently displays the highest collision rates among all methods.

Turning to aggregate throughput, as presented in Fig. 2 (right plot), DQN, marked by red diamonds and a dash-dot line, demonstrates the highest throughput across all tested node counts, consistently outperforming the other methods. Gemini2.5pro, represented by blue circles and a solid line, shows good initial throughput, but it declines as the number of nodes increases. Deepseek_rl, with orange squares and a dashed line, presents a fluctuating throughput, peaking at a moderate node count before decreasing. Lastly, DCF, denoted by green triangles and a dash-dot line, consistently exhibits the lowest aggregate throughput throughout the range of tested nodes.

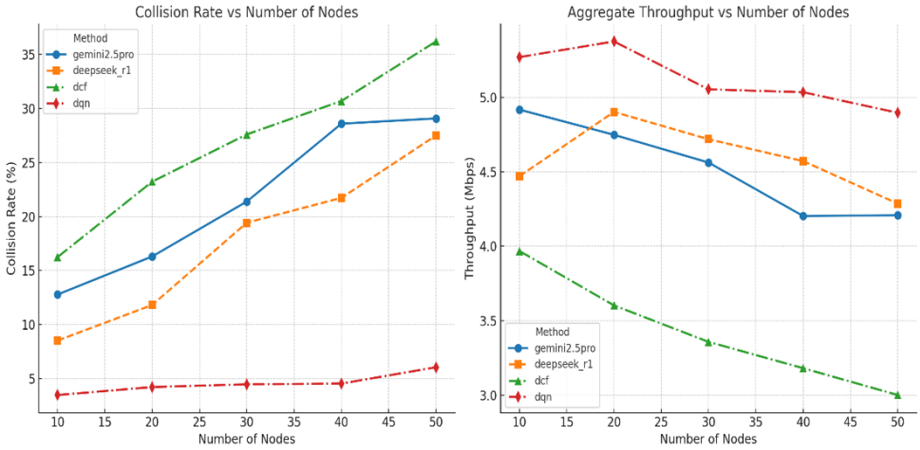

Fig. 2. Performance comparison: Collision rate and Aggregate throughput under varying numbers of nodes.

In conclusion, the data presented in Fig. 2 consistently indicates that DQN achieves superior performance compared to gemini2.5pro, deepseek_rl, and DCF. DQN effectively minimizes collision rates while simultaneously maximizing aggregate throughput across different network sizes. This dual advantage positions DQN as the most robust solution for ensuring a high Quality of Service (QoS) in the evaluated network scenarios. As previously noted in the accompanying Chinese statement, a complementary role for LLMs (potentially gemini2.5pro or deepseek_r1) is suggested. These models can provide real-time assistance to Access Points (APs), especially during periods of DQN training or network architecture adjustments. While LLMs may not match DQN's direct performance metrics, their adaptability offers a valuable mechanism for maintaining network QoS during maintenance or updates to the primary reinforcement learning agent.

4.4 Convergence Time

To thoroughly evaluate the performance of Gemini and DeepSeek (DS) under multi-node conditions, this section focuses on three key metrics: Collision Rate, Throughput, and Learning Convergence Behavior (DQN).

As shown in Fig. 3, the collision rate trends highlight the differences in contention management among methods. Gemini consistently maintains a lower collision rate compared to DS, particularly in high-density scenarios (e.g., 40 nodes). Although DS momentarily outperforms Gemini under lower node counts (e.g., 10-20 nodes), Gemini demonstrates a stronger capacity to manage concurrent transmissions and reduce interference as load increases. This advantage is crucial for sustaining stable performance in dense wireless environments. In contrast, DS exhibits a sharper rise in collision rate as node density increases, indicating less effective contention resolution.

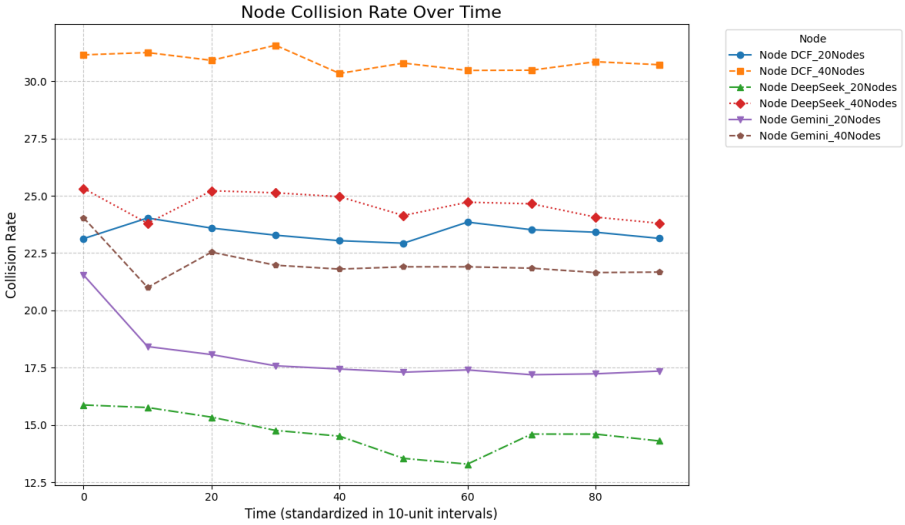

Fig. 3. Collision rate comparison over time in DCFLLM

In Fig. 4, the throughput comparison further validates Gemini's superior performance. Across both 20-node and 40-node settings, Gemini consistently delivers higher aggregate throughput than DS and DCF. This reflects its ability to make efficient use of channel resources while avoiding excessive collisions. DS, while maintaining reasonable throughput in low to moderate densities, lags behind Gemini due to its higher collision rate and more conservative access strategies.

The cumulative throughput convergence depicted in Fig. 5 focuses on DQN20 and DQN40, which represent DQN-based learning agents operating under 20 and

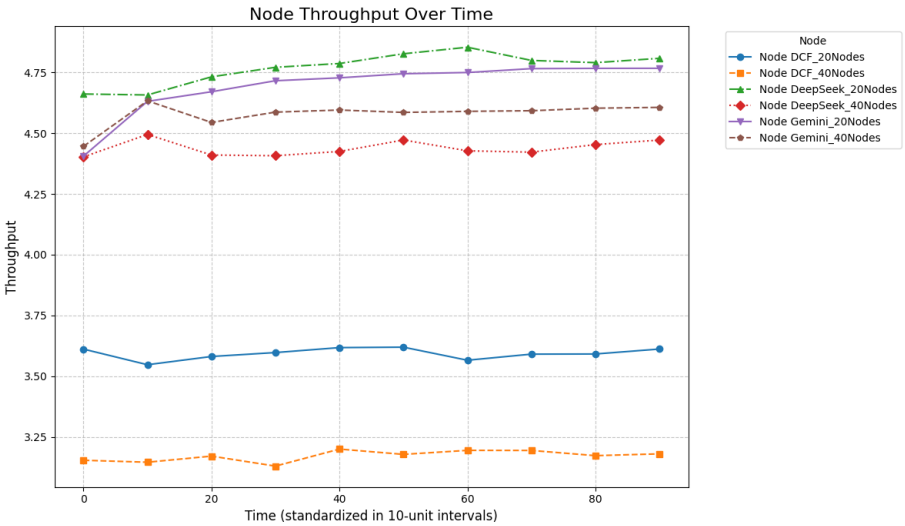

Fig. 4. Aggregate throughput comparisons over time in DCFLLM

40 nodes, respectively. Over time, these models gradually improve as they undergo training. Remarkably, Gemini powered by a pre-trained large language model (LLM)—achieves comparable or even superior throughput levels prior to the DQN models reaching convergence. Based on the trend observed, Gemini's initial performance matches that of DQN after approximately 5,000 to 7,000 seconds of training, depending on the node count. This demonstrates the capability of LLMs to offer near-optimal strategies from the outset, without requiring extensive online exploration. Such behavior is highly beneficial for time-sensitive or rapidly evolving deployments.

In summary, Gemini stands out as a robust and scalable solution for multi-node wireless networks. It effectively minimizes collisions, sustains high throughput, and offers immediate operational performance key attributes for low-latency or dynamic wireless scenarios. While DS may not match Gemini in throughput, it still offers a fair and stable alternative in applications that prioritize controlled adaptation.

5 Conclusion

While DCFLLM shows strong potential for deployment in wireless network environments, several important limitations merit consideration. First, inference latency from cloud-based models like Gemini may hinder real-time responsiveness, particularly in rapidly changing wireless conditions. Although the system partially mitigates this by applying decisions in subsequent control rounds, such latency may still impact time-sensitive applications. Second, LLMs exhibit high

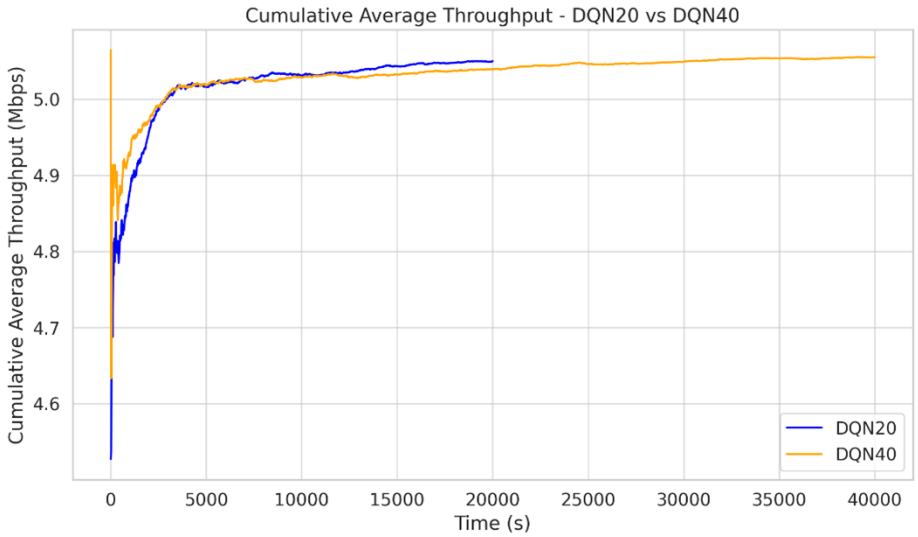

Fig. 5. Average throughput over time in DQN

sensitivity to prompt design—even minor changes in prompt structure or semantics can result in significant variations in output, posing challenges for consistent and reliable deployment. Third, despite the use of structured prompts, there remains a risk of hallucinations, where the model generates inaccurate or invalid outputs, especially when faced with ambiguous or noisy inputs. Finally, the computational cost of inference, particularly for large models like DeepSeek-30B or Gemini, may limit the feasibility of deploying DCFLLM on edge devices with constrained resources, even though model training is not required.

It is also important to note that the current study is conducted under the IEEE 802.11b standard. While this provides a controlled and well-understood environment for initial validation, it does not reflect the complexity of modern wireless systems. Future work should therefore explore extending DCFLLM to newer standards such as IEEE 802.11ax (Wi-Fi 6) or even Wi-Fi 7, which introduce more sophisticated MAC mechanisms (e.g., OFDMA, BSS coloring, and dynamic scheduling) and present additional challenges and opportunities for intelligent MAC layer adaptation.

To address these broader limitations, future work should explore the development of lightweight LLMs, robust and modular prompt templates, and model compression or distillation techniques to reduce runtime overhead while preserving decision quality. Moreover, we suggest deploying DCFLLM on real-world WLAN testbeds to evaluate its effectiveness under practical conditions. Incorporating lightweight fine-tuning approaches (e.g., LoRA) and Retrieval-Augmented Generation (RAG) can further improve context awareness and decision accuracy by grounding the model in up-to-date wireless standards and policy documentation. Finally, hybrid systems that integrate LLMs with reinforcement learning

agents—for example, using LLMs to provide warm-up guidance or serve as fallback control mechanisms—may offer a compelling tradeoff between adaptability, performance, and deployment feasibility.

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